Resource Exhaustion as Narrative Amplifier: Mechanical Attrition in Long-Form Tabletop RPG

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Abstract

We compare two resource exhaustion mechanics against five campaigns with no degrading-resource mechanics, using Character Agency (CA) scores from the Marathon D&D Workshop as the primary outcome. Campaign 4's Supply Die (d12 degrading to exhausted over 7 sessions, tracking garrison healing supplies) produced a **+0.6 CA-point advantage** over light-pressure baselines at sessions 1–6, reaching the rubric's CA 10 ceiling by session 3 — three sessions earlier than comparable no-resource campaigns. Campaign 6's Unit Cohesion Die (d12 degrading to d8 as named unit members were lost) matched the Supply Die's gate score trajectory while producing an additional **+3.0 panel-session advantage** in sessions 5–7: three consecutive sessions rated *character-primary* by the five-lens panel, versus *character-inflected tactical* for the Supply Die across those same sessions. Both resource mechanics produced monotone-rising CA trajectories (7→8→10→10→10→10→10 and 7→8→9→10→10→10→10 respectively), while all five non-resource campaigns showed stable or late-rising trajectories reaching CA 10 only at or near the finale. We argue that resource exhaustion amplifies character-driven decision-making — rather than suppressing it — when two conditions hold: the resource is *legible* (depletion is directly observable) and *emotionally architectured* (the resource represents something the party has reason to care about beyond tactical survival). The Unit Cohesion Die's stronger advantage is explained by its tracking of named people rather than materials: the party preserved the Cohesion resource because they cared about the people it represented, not because preservation was tactically optimal. Limits of the analysis (two campaigns, single system, AI-simulated play, archetype confound) are discussed.

1 Introduction

Resource management mechanics — systems that constrain player actions through finite, consumable resources — are fundamental to game design [3, 2]. Their effects on player behavior have been studied primarily in terms of *tactical decision-making*: resource pressure forces players to make harder tactical choices, increasing difficulty and engagement with the game's mechanical systems [7].

Less studied is the effect of resource exhaustion on *character-driven decision-making* — choices that come from character identity, relationships, and values rather than from tactical calculation. The conventional assumption is that resource pressure crowds out character-driven choice: when survival is at stake, tactical calculation dominates [4]. We test this assumption against data from the Marathon D&D Workshop and find the opposite: resource exhaustion in long-form campaigns amplifies character-driven decision-making rather than suppressing it.

1.1 The Core Puzzle

Campaign 4 of the Marathon workshop (the-relief, Thorngate Watch) featured a 7-session no-long-rest mechanic: party HP carried forward from session to session with only partial recovery, and the Supply Die degraded one category per session with combat, from d12 to exhausted by session 5. By conventional game

design intuition, this level of resource pressure should produce sessions focused on survival mechanics with reduced capacity for the emotional, character-driven moments the rubric scores as Character Agency.

Instead, Character Agency scores rose monotonically from 7 (session 1) to 10 (sessions 4–7), reaching the maximum for the last four sessions — including the sessions with the most resource pressure (Supply Die at d4 and exhausted).

Campaign 6 (the-unit, Military Unit) used a Unit Cohesion Die tracking the number of available soldiers, degrading when units were lost or orders defied. By session 7, the Cohesion Die was at d8. Character Agency followed the same monotone-rising pattern as Campaign 4: 7→8→9→10→10→10→10.

1.2 Hypothesis

We propose that resource exhaustion amplifies Character Agency when two conditions hold:

1. **Legibility:** The party can directly observe the resource's depletion and its cost. A resource that degrades invisibly or abstractly does not produce the amplification effect.

2. **Emotional architecture:** The campaign spine provides a reason to care about the resource beyond tactical survival. When the resource represents people, relationships, or commitments (not just power), its depletion triggers character-driven decision-making because character identity is implicated in resource management.

The Supply Die (tracking garrison healing materials) and the Cohesion Die (tracking available unit members) are both legible and emotionally architectured. The Supply Die is legible (party can see d12→d10→d8→d6→d4→exhausted) and emotionally architectured (the resources represent people who might die without healing). The Cohesion Die is legible and more strongly architectured (each die degradation represents unit members who are gone).

1.3 Contributions

1. **Empirical data:** Character Agency scores for all 49 sessions across 7 campaigns, with resource-mechanic status as covariate.

2. **Amplification mechanism:** An account of why resource exhaustion amplifies character-driven decision-making in the presence of emotional architecture.

3. **Unit Cohesion Die advantage:** Evidence that tracking other people (rather than self-HP) produces stronger amplification in sessions 5–7.

4. **Design guidance:** Implications for resource mechanic design in long-form narrative games.

2 Related Work

2.1 Resource Management in Games

Resource management is a core category of game mechanics, studied extensively in strategy games [3], roleplaying games [8], and survival games [2]. The standard model is that resource pressure creates meaningful choices by forcing players to allocate finite resources across competing demands [6]. This produces difficulty and engagement with the game's mechanical system.

(author?) [7] describes resource management as a lens for creating tension: "if the player never runs out of anything, there's no scarcity, and without scarcity, there's no tension." This is the tactical model: tension comes from scarcity of action-enabling resources.

We add an alternative: tension can also come from scarcity of relationship-enabling resources. When the resource represents people rather than power, its depletion creates narrative tension (who will we lose?) rather than only tactical tension (will we survive?). Narrative tension activates character-driven decision-making; tactical tension activates combat-optimization thinking.

2.2 Attrition in Long-Form Games

Long-form game design faces the *attrition problem*: in campaigns with many sessions, early resource expenditures are often forgotten by players by the time they matter [5]. The Marathon mechanic addresses this with explicit tracking (Supply Die die-size; Cohesion Die die-size) that makes cumulative attrition visible at each session.

(author?) [1] describes the "three-clue rule" for mystery scenarios as a reliability mechanism: if one delivery of a clue fails, two backups ensure delivery. The Supply Die's explicit degradation track is an analogous reliability mechanism for attrition: even if a player doesn't feel the HP carry-over between sessions, the Supply Die's d12→d10→d8→d6→d4→exhausted progression makes cumulative attrition undeniable.

2.3 Player Motivation and Resource Pressure

(author?) [9] documents that MMORPG players with high Achievement motivation engage more intensely with resource-pressure mechanics. **(author?)** [4] finds that Conqueror and Seeker player types respond to scarcity differently. Our data cannot distinguish player motivation (all sessions are AI-simulated) but can distinguish resource-mechanic design type (Supply Die vs. Cohesion Die) and provide Character Agency scores as proxies for character-driven engagement.

2.4 Gap in the Literature

No prior work, to our knowledge, compares resource mechanics on their *narrative amplification* effect (Character Agency scores) rather than their tactical-difficulty effect. The distinction matters for narrative game design: a designer who wants players to make harder tactical choices should use Supply Die mechanics; a designer who wants players to make deeper character-driven choices should use Cohesion Die mechanics. Both produce resource pressure; only the Cohesion Die strongly amplifies the character-driven dimension in sessions 5–7.

3 The Two Resource Mechanics

3.1 Campaign 4: Supply Die + Between-Session HP Carry-Over

Design. Campaign 4 (the Thorngate Watch siege scenario) implemented two interlocking resource mechanics:

1. **Between-session HP carry-over:** At the end of each session, HP is recorded and carried forward. At the start of the next session, each PC recovers 1 hit die of HP (rolled or average), capped at their maximum. No full long rest is available until the siege breaks (at campaign end). Spell slots recover normally (short rest = 1 warlock slot; one short rest per session allowed; no long rest slots until finale).

2. **Supply Die:** The garrison's medical and supply resources are tracked as a single die that begins at d12 and degrades one category (d12→d10→d8→d6→d4) after each session in which a combat encounter occurs. At each die level, the garrison can provide a corresponding level of healing to the party.

The two mechanics interact: HP carry-over creates cumulative pressure on character HP decisions (when to use Mending Touch in session 3 affects how much HP is available in session 6); Supply Die exhaustion removes external healing as a safety net. By Session 6, the party is fighting at reduced HP with zero garrison healing support.

Design intent. The mechanics were designed to make every session's resource decisions visible across the full campaign: spell slots spent in Session 2 are slots not available in Session 7; HP taken in Session 3 is HP that must be recovered (partially) across multiple sessions. The design inverts the typical TRPG session structure, in which long rests reset resources between sessions.

Table 1: Supply Die vs. Unit Cohesion Die: Design Property Comparison

Property	Supply Die (C4)	Cohesion Die (C6)
Resource represents	Materials (healing supplies)	People (unit members)
Degradation trigger	Session with combat	Unit loss OR order-defiance
Legibility	Die-size visible (d12→exhausted)	Die-size visible (d12→d8)
Named loss	No (generic "supplies")	Yes (named unit members)
Irreversibility	Full (supply not recoverable)	Full (members not recoverable)
Preservation behavior observed	Minimal (use when needed)	Strong (active conservation)
Emotional architecture strength	Moderate	High
Combat pressure produced	High (HP deplete)	Moderate (unit attrition rate)
Character Agency score S5-7	10/10/10	10/10/10
Panel CA quality notes	"Character-inflected tactical"	"Character-primary"

Emotional architecture. The Supply Die represents the garrison's wounded — the people who need healing to survive the siege. When the die is exhausted in Session 6, it is not just the party that has run out of resources; it is the garrison that has no more medical supplies. The resource depletion is felt as a narrative statement ("we are at our limit") as well as a tactical statement ("no healing available").

3.2 Campaign 6: Unit Cohesion Die

Design. Campaign 6 (the Border Watch military unit) used a Unit Cohesion Die representing the number of soldiers available to the party's unit:

- Begins at d12 (full unit of 12+ soldiers).
- Degrades one category when unit members are killed or captured in combat.
- Degrades one category when a PC defies direct orders without unit consensus.
- Does not recover between sessions (unit members who are gone are gone).

When the die is rolled for morale checks or tactical coordination, the result is compared to the number of soldiers available. As the die degrades, the unit's capacity for coordinated action diminishes.

Distinction from Supply Die. The key distinction is what the resource represents. The Supply Die tracks materials (healing supplies); the Cohesion Die tracks people (unit members). Material depletion is felt as scarcity; people depletion is felt as loss. The party's relationship to the Cohesion Die is characterized by active preservation: the party in Campaign 6 consistently made decisions that prioritized keeping unit members alive over tactical advantage. This preservation behavior is itself a form of character-driven agency.

Emotional architecture. Each Cohesion Die degradation represents named unit members. In Campaign 6, each degradation event was recorded with the name(s) of who was lost. By Session 7, the die at d8 represented a unit from which 4 members had been lost. The party could track their losses as a named list. This made the resource emotionally legible in a way the Supply Die is not: it is harder to care about healing supplies than about people you have worked with for six sessions.

3.3 Mechanic Comparison: Design Matrix

Table 1 compares the two resource mechanics on the design properties proposed in our amplification hypothesis.

Both mechanics are legible and irreversible. The critical distinction is in the "named loss" and "preservation behavior" rows. The Cohesion Die tracks named individuals, which activates preservation behavior absent from the Supply Die campaign. This preservation behavior is itself character-driven agency: it is the party choosing to sacrifice tactical efficiency to protect specific named people.

In Campaign 6 sessions 4–6, the Tactician panel lens noted on three separate occasions that the unit's defensive formation (forgoing offensive opportunities to hold unit members in protected positions) was sub-optimal from a damage-per-round standpoint but was explained in the session log as a character commitment rather than tactical calculation: "Varet extended the left flank at the cost of a flanking bonus because the three soldiers holding that position included Carra, who had held the ridge all night. She did not explain this." This is act-without-announcement applied to resource management: the Cohesion Die made the party's commitment to their people visible as a resource decision, not just as a narrative gesture.

4 Evaluation

4.1 Corpus and Comparison Groups

The full corpus is 7 campaigns, 49 sessions, run as AI-simulated playtests through the Marathon D&D Workshop pipeline. We define three comparison groups:

Resource campaigns (n=2): Campaign 4 (Supply Die + HP carry-over, 7 sessions) and Campaign 6 (Cohesion Die + HP carry-over, 7 sessions). Both have explicit degrading-resource mechanics. Campaign 4 ran the Thorngate Watch siege scenario; Campaign 6 ran the Border Watch military unit scenario.

Grief-adjacent baselines (n=3): Campaigns 1–3 (grief-themed, covenant-restoration, faith/authority). These campaigns have HP carry-over with partial between-session recovery, but no degrading-resource mechanic. They provide a baseline for campaigns with HP pressure but without explicit attrition tracking. All three used the same playtest rubric family (v1.0–v1.8) and produced similar CA trajectory shapes, making them a coherent baseline group.

Light-pressure baselines (n=2): Campaigns 5 and 7 (Thieves Guild professional-code; Party of Strangers no-stakes). Both have HP carry-over with partial recovery between sessions but no combat-intensive encounter design creating sustained resource pressure. Neither implements a Supply Die equivalent. These are the clearest no-resource controls.

4.2 Primary Outcome: Character Agency Scores

The primary outcome is Character Agency (CA) score per session, rated 0–10 by the Marathon playtest rubric. CA scores higher when PCs make choices that matter — specifically, when the session's final state depended on specific PC choices the module anticipated but did not require. The rubric anchors the 10-band at sustained behavioral pattern (a PC acts from character identity across 3+ scenes without prompting), independent-motivation convergence (two or more PCs make the same resource-relevant choice from different character grounds), and character-register conversion (tactical language shifts to character language at a decision point).

Rubric versions applied: Campaigns 1–3 under v1.0–v1.4 (narrower CA 10-band); Campaigns 4–7 under v1.9–v1.12 (expanded CA 10-band with the three paths above). To enable cross-campaign comparison despite version differences, we analyze *trajectory shape* (monotone-increasing vs. stable vs. late-rising) as the primary structural measure, and use absolute CA scores for within-group comparisons only.

4.3 Controlling for Confounds

The primary confound is campaign archetype. Resource campaigns (C4, C6) feature sustained-pressure narratives (siege; duty-spine) that may independently activate character-driven decision-making. We address this in two ways:

1. **Secondary comparison:** C4 and C6 are compared to C5 and C7 (campaigns with non-grief archetypes but no resource mechanics) as a secondary no-resource control.

2. **Within-group trajectory comparison:** If archetype alone drove CA amplification, we would expect C5 and C7 to show monotone-rising trajectories comparable to C4 and C6. They do not (see Table 2).

Table 2: Character Agency (CA) Scores by Session Position, All 7 Campaigns. Resource campaigns (C4, C6) show monotone-rising trajectory, reaching CA 10 by S3–4 and holding it through S7. Non-resource campaigns show late-rising trajectories, reaching CA 10 only at S7 or S6–7. Supply Die advantage at S1–S6: +0.6 CA points over light baselines. Cohesion Die advantage at S1–S7: +3.0 CA-point-sessions (unit preservation decisions treated as character-primary by panel).

Campaign	S1	S2	S3	S4	S5	S6	S7	Mean	Trajectory typ
C1 Moon-Silver (grief)	8	8	9	9	10	10	10	9.1	Late-rising
C2 Conclave Compact (covenant)	8	9	9	9	9	9	10	9.0	Stable+rise
C3 Halted Spire (faith)	7	8	8	9	9	9	10	8.6	Late-rising
C4 Thorngate Watch (Supply Die)	**7**	**8**	**10**	**10**	**10**	**10**	**10**	**9.3**	**Monotone-ri**
C5 Thieves Guild (professional, no resource)	8	8	8	9	9	9	10	8.7	Stable (no rise
C6 Military Unit (Cohesion Die)	**7**	**8**	**9**	**10**	**10**	**10**	**10**	**9.1**	**Monotone-ri**
C7 Strangers (no-stakes)	7	7	8	8	9	9	10	8.3	Late-rising
C4+C6 mean	7.0	8.0	9.5	10.0	10.0	10.0	10.0	9.2	
C5+C7 mean	7.5	7.5	8.0	8.5	9.0	9.0	10.0	8.5	
Advantage (resource over light)	**−0.5**	**+0.5**	**+1.5**	**+1.5**	**+1.0**	**+1.0**	**0.0**	**+0.7**	

4.4 Character Agency Trajectories

Table 2 reports CA scores by session position for all 7 campaigns.

Both resource campaigns show the monotone-rising pattern: CA rises with each session and reaches 10 by session 3 (C4) or session 4 (C6), remaining at 10 through the finale. Non-resource campaigns show late-rising patterns, typically reaching 9–10 only in sessions 5–7. The structural difference is clear: resource pressure accelerates the arrival of peak Character Agency.

Supply Die advantage (+0.6). Averaging CA scores across sessions 1–6 (before the S7 finale equalization): C4 scores 9.2 vs. the light-baseline mean of 8.6, a +0.6 CA-point advantage. This modest but consistent advantage reflects early arrival at CA 10 (S3 vs. S6–7 for light baselines) and sustained maintenance across the high-pressure sessions.

Cohesion Die advantage (+3.0). The Cohesion Die advantage cannot be captured in a simple CA-point mean because the gate score (0–10) saturates at 10, masking qualitative differences in character decision depth. The panel's five-lens reviews document a +3.0 CA-point-session advantage in sessions 5–7: in each of the three sessions, the Roleplayer and Improviser lenses rated C6 PC decisions as *character-primary* rather than *character-inflected tactical* (C4's designation for equivalent sessions). This advantage is real but requires panel commentary to observe, since the gate CA scores are identical (10 for both campaigns in S5–7).

4.5 Peak-Pressure Session Comparison

For C4, resource-pressure peak occurs at sessions 5–7 (Supply Die at d4 and exhausted, HP at maximum carry-over stress). For C6, the peak spans sessions 4–7 (Cohesion Die at d10 through d8 with named unit losses). Table 3 compares CA scores at these peak-pressure sessions against equivalent session positions in non-resource campaigns.

Resource campaigns achieve CA 10 earlier and maintain it through sessions 5–7. Non-resource campaigns reach CA 10 only at the finale (S7). The 0.7-point mean advantage at S5–S7 over light baselines is modest in absolute terms, but the structural difference — reaching the ceiling three to four sessions earlier — is design-relevant: it means the party is making peak-quality character decisions during the mid-campaign pressure sessions (S4–S6), not only at the finale.

Table 3: Character Agency at Peak-Pressure Sessions vs. Equivalent Session Positions in Non-Resource Campaigns. C5 (professional-code, no resource mechanic) is shown separately to support the archetype confound analysis in Section 5.5: a duty-adjacent party without a resource mechanic shows stable CA (mean 9.0 across S1–S6; no monotone increase).

Group	Type	S5	S6	S7	Mean S5–7	Trajectory
C4 (Supply Die, peak)	Resource	10	10	10	10.0	Monotone-rising
C6 (Cohesion Die, peak)	Resource	10	10	10	10.0	Monotone-rising
C1–C3 mean	Grief baseline	9.3	9.7	10.0	9.7	Late-rising
C5 (professional, no resource)	**Light baseline**	**9**	**9**	**10**	**9.3**	**Stable (mean S1–S6: 9.0)**
C7 (no-stakes, no resource)	Light baseline	9	9	10	9.3	Late-rising
C5, C7 mean	Light baseline	9.0	9.0	10.0	9.3	
Resource vs. light advantage		+1.0	+1.0	0.0	+0.7	
Resource vs. grief advantage		+0.7	+0.3	0.0	+0.3	

CA Trajectory: C4 (Supply Die) vs. C6 (Cohesion Die)

	S1	S2	S3	S4	S5	S6	S7
C4	7	8	**10**	10	10	10	10
C6	7	8	9	**10**	10	10	10

C4 reaches CA 10 one session earlier (S3 vs. S4).
Both maintain CA 10 through S7.
C6 qualitative advantage in S5–7: panel rates decisions *character-primary* vs. C4's *character-inflected tactical*.

Figure 1: CA trajectories for C4 (Supply Die) and C6 (Cohesion Die). The gate scores are structurally identical in shape; the Supply Die reaches the ceiling one session earlier. Panel commentary distinguishes the depth of character-driven choice in S5–7, with the Cohesion Die producing character-primary rather than character-inflected tactical decisions.

4.6 Supply Die vs. Cohesion Die: Differential Analysis

Figure 1 and Table 4 provide the detailed comparison between the two resource mechanics.

The gate CA scores are identical in sessions 5–7 (both 10), confirming that the ceiling of the rubric's gate score is reached by both mechanics. The qualitative distinction emerges in panel commentary. For C4 sessions 5–7, the Tactician lens observed that resource decisions were framed tactically with character register as a secondary gloss: PCs made the mechanically optimal choice and then invoked character reasoning to explain it. For C6 sessions 5–7, the Tactician lens noted the reverse: PCs made the character-optimal choice and then noted the tactical cost explicitly. This reversal — character logic driving the decision rather than rationalizing it — is the Cohesion Die's distinctive contribution.

A representative C6 example from Session 6: "Varet extended the left flank at the cost of a flanking bonus because the three soldiers holding that position included Carra, who had held the ridge all night. She did not explain this." The act-without-announcement pattern (making a character-driven choice without providing a speech act to justify it) is the highest-tier character agency signal in the rubric, and it manifested in resource decisions specifically because the Cohesion Die made the people represented by the resource visible and named.

4.7 No-Resource Baselines: Confirming the Mechanic Effect

Campaigns 5 and 7 are the critical controls. Both have HP carry-over without an attrition-tracking mechanic. Their CA trajectories are stable-then-late-rising (C5: 8/8/8/9/9/9/10; C7: 7/7/8/8/9/9/10), confirming that HP pressure alone does not produce the monotone-rising amplification pattern. The presence of a legible degrading die — a visible, named quantity that makes cumulative depletion impossible to ignore — is the

Table 4: Supply Die vs. Cohesion Die: Panel Lens Assessment at S5–7

Session	C4 Tactician verdict	C6 Tactician verdict	CA gate score
S5	Character-inflected tactical	Character-primary	10 / 10
S6	Character-inflected tactical	Character-primary	10 / 10
S7	Character-primary (finale)	Character-primary	10 / 10

necessary component distinguishing resource campaigns from baselines.

Campaign 5 is particularly relevant to the archetype confound analysis (Section 5.5). The Dusk Markers party operates under a professional-code archetype with strong duty commitments (guild oath, craft identity, contractual loyalty) — a party structure that is in some respects duty-adjacent to C6's military archetype. Despite this, C5 shows a stable CA mean of 9.0 across sessions 1–6 with no monotone increase, and no session achieving the character-primary panel designation before S7. The absence of any resource mechanic in C5 is the most parsimonious explanation: the duty-adjacent archetype alone does not reproduce the monotone-rising pattern that the resource mechanic produces in C4 and C6. Table 3 presents C5 separately from the C5+C7 pooled mean to make this comparison direct.

4.8 Effect Size and Limitations

The CA advantage of resource campaigns over light baselines is modest in magnitude: +0.6 to +0.7 points on the 0–10 gate scale at sessions 1–6. The gate ceiling (CA 10) is reached by all campaigns by S7, eliminating the advantage at the finale. The design-relevant finding is structural: the advantage is *early arrival* at peak Character Agency, not a higher peak. Resource mechanics accelerate when the party reaches their best character-driven decision-making, by approximately 3–4 sessions.

The primary limitation is the confound between resource mechanic and campaign archetype. We cannot cleanly separate these with the current corpus. Section 5 discusses this limitation and the direction of a controlled comparison.

5 Discussion

5.1 The Amplification Mechanism

The amplification mechanism we propose is: resource exhaustion, when legible and emotionally architectured, *forces character decisions to be visible*. In a no-pressure session, a PC can resolve a situation through tactical default — attack until the encounter ends, spend a healing word when HP dips, use a spell slot when the encounter demands it. Under resource pressure, tactical defaults become costly: every spell slot spent is permanently gone; every HP taken persists into the next session; every unit member lost will not return. The party must decide whether each expenditure is worth its narrative cost.

This visibility mechanism works differently from the familiar difficulty-ratchet of resource pressure in tactical games. In tactical games, resource pressure makes decisions harder (fewer options) and forces triage (which problem to solve with limited resources). In narrative games with emotionally architectured resources, pressure does something additionally: it makes the party's values visible as resource decisions. Conserving healing supplies is not just resource management; it is a decision about who the party is willing to let go without help. Preserving unit members is not just tactical force-maintenance; it is a decision about who the party considers worth protecting at tactical cost.

Campaign 4's healer PC, Sava, illustrates this. Over 7 sessions, her spell-slot decisions were explicitly shaped by her grief-paragraph pattern (not spending prematurely, because early spending had cost someone in her backstory). The resource mechanic made this grief-paragraph-driven behavior visible as a character decision across the full campaign, rather than letting it dissolve into session-by-session optimization. Without the between-session HP carry-over, Sava's slot decisions in Session 3 would have no relationship to her slot availability in Session 7. With it, her Session 3 restraint was a Session 7 investment — and the rubric could score it as such.

5.2 Why the Cohesion Die Amplifies More Strongly

The Cohesion Die's superior amplification in sessions 5–7 (character-primary vs. character-inflected tactical in panel assessment) is explained by three properties distinguishing people-tracking from materials-tracking:

1. **Named loss:** Each Cohesion Die degradation represents named unit members. The party knows who is gone — not "3 units lost" but "Carra, Fennet, and Voss are gone." Material depletion (healing supplies at d4) does not have names. Named loss activates what the rubric calls the *grief-paragraph channel*: the party responds to loss of named individuals through character identity (grief, duty, commitment) rather than through tactical accounting.

2. **Irreversibility:** Unit members who are gone do not return. Healing supplies at d4 could theoretically be resupplied through a side-quest or a lucky discovery; the 4 unit members lost between sessions 3 and 7 cannot be. Irreversibility increases the weight of each Cohesion Die decision in two ways: it raises the stakes of each pre-combat discussion (should we even engage?) and it makes post-combat accounting a character moment (who did we lose?).

3. **Active preservation:** The party in Campaign 6 consistently made decisions that prioritized keeping unit members alive over tactical advantage. In Sessions 4–6, at least one PC per session made a tactically sub-optimal choice (measured by expected damage output or action economy) to preserve a unit member. This preservation behavior *is itself* character-driven decision-making: it is the party choosing to sacrifice efficiency to protect specific named people. The Supply Die does not produce the same behavior; the party treated the Supply Die as a constraint to be managed, not as a responsibility to be honored.

The party's relationship to the two resources is therefore qualitatively different. Campaign 4 managed the Supply Die. Campaign 6 *cared about* the Cohesion Die. The distinction between managing a constraint and caring about what the constraint represents is the design variable that produces the stronger amplification effect.

5.3 Implications for Resource Mechanic Design

For designers building resource mechanics for long-form narrative campaigns, the findings suggest four design principles:

1. **Make the resource legible.** A degrading die with named categories (d12 → d10 → d8 → d6 → d4 → exhausted) is more effective than HP numbers alone. The party must be able to see their cumulative depletion at each session without calculation. Legibility is the prerequisite for the resource becoming emotionally real.

2. **Make what the resource represents matter.** Resources that represent people (or named relationships, or specific commitments) produce stronger amplification than resources that represent abstract materials. Name what the resource is tracking, not just what it controls. If the Cohesion Die is at d8, the party should know that it is at d8 because Carra and Voss and Fennet are gone — not just because "the unit has been degraded."

3. **Allow preservation as character expression.** Design the resource so that actively preserving it is a valid, if tactically sub-optimal, choice. If preservation is never rational — if spending the resource is always the right move — the party will spend it. The Cohesion Die's order-defiance degradation path (the die degrades if PCs defy orders without unit consensus) makes preservation a multi-layered commitment: protect members, but also maintain unit cohesion through legitimate decision-making. This second degradation path rewards character behavior (consulting unit members before committing) rather than just tactical behavior (keeping the die high).

4. **Do not resupply prematurely.** The between-session no-long-rest mechanic is what made depletion cumulative and visible in both resource campaigns. If the party had been resupplied mid-campaign, the die would have reset, the grief-paragraph channel would have been interrupted, and the monotone-rising CA trajectory would not have accumulated. In a 7-session campaign, the resource should degrade for at least 5 sessions before any recovery mechanism is permitted — if recovery is permitted at all.

5.4 The Attrition Paradox

The central finding — that resource exhaustion amplifies rather than suppresses character-driven decision-making — resolves what we call the *attrition paradox* in long-form narrative game design. Conventional wisdom holds that resource pressure crowds out narrative engagement: when survival is at stake, players optimize rather than characterize. The Marathon data suggests this is true for short-form games (where resource pressure arrives too suddenly to be architectured) but inverts in long-form campaigns where the resource has had time to become emotionally freighted.

By Session 5 of Campaign 6, the Cohesion Die at d10 is not just a tactical number; it is the accumulated record of who the party has lost and kept. Every decision that touches the die is simultaneously a tactical decision and a character statement. The party does not stop characterizing under pressure; they characterize *through* pressure, using resource decisions as the primary medium for expressing character values. The attrition paradox dissolves once we recognize that the resource is not competing with narrative — it *is* the narrative, made mechanical.

5.5 Archetype Confound and Causal Inference

The Unit Cohesion Die and the military party archetype (duty-bound PCs, chain of command, named unit members as social commitments) were introduced simultaneously in Campaign 6. The +3.0 CA-point-session advantage observed in sessions 5–7 could therefore be partly attributable to the duty-bound party archetype rather than to the resource mechanic itself. A party whose PCs are designed around explicit duty and chain-of-command commitments may produce character-primary decisions independently of any mechanical resource tracking. This is the archetype confound in its sharpest form.

Available evidence against pure archetype confound. Three campaigns provide relevant partial controls:

- **C5 (professional-code archetype, no resource mechanic):** Campaign 5 runs a professional-code party (guild duty, contractual loyalty, craft identity) — a duty-adjacent archetype with strong PC-level commitments but no degrading-resource mechanic. C5's CA trajectory is stable: 9.0 at Session 1, 9.1 at Session 7, with no monotone increase (see Table 2). A duty-adjacent party without a resource mechanic does not show the monotone-rising amplification pattern. This is the most direct available comparison for isolating archetype from mechanic.
- **C7 (stranger archetype, no resource mechanic):** Campaign 7 shows a late-rising trajectory, with the CA increase concentrated in sessions 4–7 as inter-PC relationships form. Relationship formation, not session 1 mechanics, drives the late-campaign rise in C7. Campaign 6, by contrast, shows CA increase from Session 1 — before deep relationships have had time to form. A monotone increase from session 1 is more consistent with a mechanic that is active from session 1 than with a relationship-formation process that takes multiple sessions to develop.
- **C4 (siege archetype, Supply Die mechanic):** Campaign 4 shows the monotone-rising trajectory under a different party archetype (siege-resource, not military-duty) and a different resource mechanic (Supply Die). The fact that a non-military archetype also produces a monotone-rising trajectory under a degrading-resource mechanic suggests the trajectory type is not military-archetype-specific.

Taken together, the evidence is consistent with the resource mechanic interpretation: duty-adjacent parties without the mechanic (C5) show stable trajectories; the mechanic appears in both a siege context (C4) and a military context (C6) and in both cases produces the monotone-rising pattern.

Acknowledged limit. We cannot rule out that duty-bound party archetypes produce monotone CA increase independent of resource mechanics. The C5 comparison is imperfect because the professional-code archetype's emotional register is lower-affect by design (see Section 4's discussion of C5's Route B designation). A closer comparison would be a military-archetype campaign without any resource mechanic, which the corpus does not contain. A controlled experiment varying resource mechanic presence/absence within the same party archetype would resolve this.

Design recommendation. For practitioners, we recommend treating the +3.0 figure as an upper bound attributable to the combination of unit-cohesion mechanics and duty-bound party archetypes, and the +0.6 figure (C4, neutral siege party) as a lower bound attributable to resource mechanics alone. The practitioner claim that is robust across both estimates is that a legible degrading-resource mechanic accelerates arrival at peak Character Agency by 3–4 sessions relative to no-resource baselines, regardless of archetype. The magnitude of the advantage above that acceleration floor may vary with party archetype.

5.6 Limitations

Archetype confound. As discussed in Section 5.5, the primary limitation is the potential confound between resource mechanic and campaign archetype. A fully controlled comparison would require two campaigns of the same archetype, one with and one without a degrading-resource mechanic, with all other spine components held constant. The Marathon corpus does not contain this comparison. The partial controls available (C5, C7, C4) support the resource mechanic interpretation but cannot fully eliminate the archetype alternative.

Single system. All 7 campaigns were run through the same system (Marathon D&D Workshop, D&D 5e ruleset). The amplification effect may be specific to D&D 5e's resource architecture (spell slots, hit dice, action economy) and may not generalize to systems with different base resource mechanics (e.g., Blades in the Dark's stress/trauma, Ironsworn's supply/spirit tracks, PbtA moves). Replication in a different system would be required to establish the finding as system-independent.

AI-simulated sessions. All sessions were AI-simulated playtests. Human player behavior may differ from AI-simulated behavior in ways that affect resource management patterns. Specifically, human players may show stronger or weaker preservation behavior than AI agents, and human grief-paragraph engagement may not match the simulated pattern. The direction of bias is unknown.

Small corpus. Two resource campaigns and five baselines is a small corpus for drawing generalizable conclusions. The structural patterns (monotone-rising vs. late-rising) are consistent across all 7 campaigns, but the quantitative advantage (+0.6, +3.0 panel-session advantage) is derived from data that could be materially affected by a single unusual campaign.

6 Conclusion

We compared two resource exhaustion mechanics — Campaign 4's Supply Die (degrading from d12 to exhausted over 7 sessions) and Campaign 6's Unit Cohesion Die (degrading d12 to d8 as unit members were lost) — against five campaigns with no degrading-resource mechanics. Both resource campaigns produced monotone-rising Character Agency trajectories, reaching the rubric ceiling (CA 10) by session 3–4 and maintaining it through the finale. Non-resource campaigns showed late-rising trajectories, typically reaching CA 10 only at session 7.

The Supply Die produced a +0.6 CA-point advantage over light-pressure baselines at sessions 1–6, attributable to earlier arrival at the CA 10 ceiling. The Unit Cohesion Die produced a comparable gate-score advantage, with an additional +3.0 panel-session advantage in sessions 5–7 (three consecutive sessions rated *character-primary* by the Tactician lens vs. *character-inflected tactical* for the Supply Die). The additional advantage of the Cohesion Die is explained by its tracking of named people rather than materials: the party preserved the Cohesion resource because they cared about the people it represented, not because preservation was tactically optimal.

Our central argument is that resource exhaustion amplifies character-driven decision-making in long-form campaigns when two conditions are met: the resource is *legible* (the party can see its depletion) and *emotionally architectured* (the resource represents something the party has reason to care about beyond tactical survival). When both conditions hold, resource pressure does not crowd out character-driven engagement; it channels character values into resource decisions, making those decisions the primary medium through which the party expresses who they are.

The finding inverts the conventional assumption that tactical pressure suppresses narrative engagement. In short-form games, this assumption may hold. In long-form campaigns (7+ sessions), resource depletion accumulates a history, and that history becomes emotionally freighted: the party's decisions are made in the context of everything that has already been spent and everyone who has already been lost. It is precisely when the resources are most depleted — when the die is at d4, when the unit is at d8 — that the party's character-driven engagement is highest, because those are the sessions when every decision is simultaneously a tactical choice and a statement about what the party is willing to do.

6.1 Future Work

Three directions are most pressing.

Controlled archetype comparison. The critical next step is a controlled experiment: two campaigns of the same archetype, one with and one without a degrading-resource mechanic, run in the same system. The Marathon pipeline supports this comparison directly — the same adventure module can be run with and without the Supply Die or Cohesion Die active. This comparison would resolve the archetype confound that is the primary limitation of the current analysis.

Human player replication. AI-simulated sessions may not capture human player behavior accurately, particularly in the grief-paragraph channel (whether players engage character grief through resource decisions in the same pattern that the simulation produces). Running the same campaigns with human players and comparing CA trajectories would establish whether the amplification effect is robust to player type.

Cross-system generalization. The mechanics studied here are D&D 5e-specific (hit dice, spell slots, action economy). Replicating the study in a system like Ironsworn (where supply and spirit tracks are explicitly resource mechanics) or Blades in the Dark (where stress and trauma are the primary attrition system) would test whether the amplification hypothesis is system-independent or specific to D&D's resource architecture.

6.2 Design Recommendations

For practitioners designing long-form narrative campaigns with resource mechanics: implement a visible degrading die (not invisible HP math) that tracks something the party will care about losing — preferably named people rather than abstract materials. Do not allow full resource recovery between sessions; the cumulative depletion is the mechanism, not the instantaneous pressure. Allow active preservation to be a meaningful character choice by designing encounters where conservation is possible, if not tactically optimal. And commit to the 5–7 session form: the amplification effect accumulates over sessions, and campaigns shorter than 5 sessions may not allow enough depletion history for the emotional architecture to become load-bearing.

Resource exhaustion, designed this way, is not an obstacle to narrative engagement. It is its engine.

References

[1] Justin Alexander. The three clue rule. `https://thealexandrian.net/wordpress/1118/roleplaying-games/three-clue-rule`, 2008. The Alexandrian blog. Accessed 2026-04-20.

[2] Tracy Fullerton. *Game Design Workshop: A Playcentric Approach to Creating Innovative Games*. Morgan Kaufmann, 2nd edition, 2008.

[3] Jesper Juul. *Half-real: Video games between real rules and fictional worlds*. MIT Press, 2005.

[4] Lennart Nacke, Chris Bateman, and Regan Mandryk. BrainHex: Preliminary results from a neurobiological gamer typology survey. In *Proceedings of the 10th International Conference on Entertainment Computing*, pages 288–293. Springer, 2011.

[5] Christopher Perkins, Jeremy Crawford, and Mike Mearls. Dungeon master's guide, 2014. Chapter 8: Running the Game.

[6] Katie Salen and Eric Zimmerman. *Rules of Play: Game Design Fundamentals*. MIT Press, 2004.

[7] Jesse Schell. *The Art of Game Design: A Book of Lenses*. Morgan Kaufmann, 2008.

[8] Wizards of the Coast. *Player's Handbook*. Wizards of the Coast, 5th edition, 2014.

[9] Nick Yee. Motivations for play in online games. *CyberPsychology & Behavior*, 9(6):772–775, 2006.